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Statistics in the High School Mathematics Curriculum

Building Sound Reasoning under Uncertain Conditions

Youngsters and adults alike are confronted daily with situations involving statistical information. Making sense of data and dealing with uncertainty are skills essential to being a wise consumer, an enlightened citizen, and an effective worker or leader in our data-driven society. The importance of statistics education as an integral part of the mathematics curriculum was signaled by NCTM in its *Curriculum and Evaluation Standards for School Mathematics* (1989). Subsequent recommendations in state frameworks, NCTM's *Principles and Standards for School Mathematics* (2000), and most recently the College Board's *Standards for College Success: Mathematics and Statistics* reflect consistent attention to data analysis, probability, and statistics across the grades (2006).

In 2007, the American Statistical Association released a report titled *Guidelines for Assessment and Instruction in Statistics Education (GAISE) Report*. That report provides learning trajectories for key ideas of statistics organized into three developmental levels—A, B, and C. Although these three levels may parallel the standard grade-level bands (elementary, middle, and secondary), they are based on students' prior statistical experiences rather than on grade level. The three companion articles in this month's issues of *Teaching Children Mathematics, Mathematics Teaching in the Middle School,* and the *Mathematics Teacher* illustrate how the GAISE report can be used to shape a coherent development of basic ideas related to the distribution of a data-based variable beginning with exploring distributions of data, then developing an understanding of center and spread, and finally building sound reasoning under uncertain conditions.

This department provides a forum to stimulate discussion on contemporary curricular issues across a K-12 audience. NCTM plans to publish sets of three articles, focused on a single curriculum issue. Each article will address the issue from the perspective of the audience of the journal in which it appears. Collectively, the articles are intended to increase communication and dialogue on issues of common interest related to curriculum. Manuscripts on any contemporary curriculum issues are welcome. Submissions can be for one article for one particular journal, or they can be for a series of three articles, one for each journal. Submit manuscripts at the appropriate Web site: tcm.msubmit.net, mtms.msubmit.net, or **mt.msubmit.net**, or contact editors Barbara Reys (reysb@missouri.edu) for *TCM*, Glenda Lappan (glappan@math.msu.edu) for *MTMS*, or Chris Hirsch (christian.hirsch@wmich.edu) for *MT*. For a perspective on K-12 statistics education, read all three articles in the series.

Statistics is the key to decision making in the information age. The importance of statistical thinking for life and work is reflected in state and provincial mathematics frameworks and in national standards. These documents all place statistics (along with number, algebra, and geometry) as one of the "big four" strands in the mathematical sciences. But statistics, often described as data analysis, is the strand least emphasized, for many reasons—lack of teacher preparation on content and pedagogy, lack of good instructional materials, lack of emphasis on standardized tests, and lack of models for integrating statistical thinking into the school mathematics curriculum.

To provide guidance on what and how to teach statistics effectively as part of the K–12 mathematics curriculum, the American Statistical Association (ASA) developed *Guidelines for Assessment and Instruction in Statistics Educa-*

tion (GAISE). This article, along with companion articles in *Teaching Children Mathematics* (Franklin and Mewborn 2008) and in *Mathematics Teaching in the Middle School* (Kader and Mamer 2008), illustrate how GAISE can be used to create a K–12 trajectory of the development of basic ideas related to the distribution of a variable based on data. These ideas are built by viewing statistics as an investigative process and by incorporating the guiding principles for teaching outlined in **figure 1**.

Distributions summarizing quantitative information are characterized through the concepts of *shape*, *center*, and *spread*. The topics covered in this article assume that students will have been exposed to the calculation and interpretation of basic measures of center (e.g., median as the middle value and mean as a fair share or balance point) and spread (e.g., interquartile range and standard deviation) and how uses of these measures relate to the shape of a distribution and the purpose of a study. The specific goal of this article (paralleling level C of the GAISE report) is to discuss the teaching of distributions of commonly used sample statistics, such as means and proportions, and how these distributions are used in inferential (i.e., plausible) reasoning. These statistics can be derived from data collected through a survey with random sampling or a randomized experiment.

Steps in the statistical problem-solving process

- Formulate questions
- Collect data
- Analyze data
- Interpret results

Guiding principles for teaching statistics

- Conceptual understanding takes precedence over procedural skill.
- Active learning is key to the development of conceptual understanding.
- Real-world data must be used wherever possible in statistics education.
- Appropriate technology is essential in order to emphasize concepts over calculations.
- All four steps of the investigative problem-solving process should be encountered at each grade level.
- The illustrative investigations should show situations in which the statistics is essential to answering a question, not just an add-on.
- Such investigations should be tied to the mathematics they illustrate, motivate, and emphasize.

Source: GAISE Report

Fig. 1 Steps in the statistical process and guiding principles for teaching statistics

Table 1

Summary of Data on the Curfew Question

	Had a Curfew	Did Not Have a Curfew	Total
Females	33	35	68
Males	20	12	32
Total	53	47	100

A SAMPLE SURVEY

One of the most straightforward, popular, and potentially interesting ways to introduce basic concepts of inferential reasoning is through a sample survey. Students should be cautioned, however, that a good sample survey is based on a random sample from a well-defined population, which is easy to conceptualize but difficult to achieve in practice. The following example brings to light the key ideas of the statistical problem-solving process in the context of collecting and analyzing data from a sample survey.

Asking a Statistics Question

Students in a high school mathematics class decided to study the strictness of parents or guardians of students in the school. They realized that, because the meaning of *strict* is ambiguous, they could not simply ask as a survey question, "Are your parents or guardians strict?" After some debate, they decided on a variety of survey questions that might get at the idea of strictness. One of these was, "Do you have a curfew?" from which they formulated the following statistics question: "What percentage of students in the school have a curfew?"

Collecting and Analyzing Appropriate Data

A random sample of 100 students was selected from the approximately 1200 students in the school, and all sampled students responded to the survey. The results (**table 1**) showed that 53 of the 100 reported having a curfew. It is important to note that data collected through this survey question are categorical (yes meant that the student did have a curfew; no meant that the student did not). However, the sample summary statistic (proportion in sample with a curfew = 0.53) is numerical.

Interpreting Results

Students should understand that the true proportion of all students in the school is not likely to be 0.53 and that another sample of 100 students might very well produce a different result. This is the underlying idea of a sampling distribution: The sample proportion will vary from one sample to another. How much variation can we expect? Time for technology! If we assume that the true population proportion is 0.53, students can simulate the repeated selection of samples of size 100 from a "theoretical" population that has a 53 percent chance of generating a favorable outcome. An easy way to do this is to sample 100 two-digit random numbers, with the numbers 01 through 53 representing yes and the numbers 54 through 99 as well

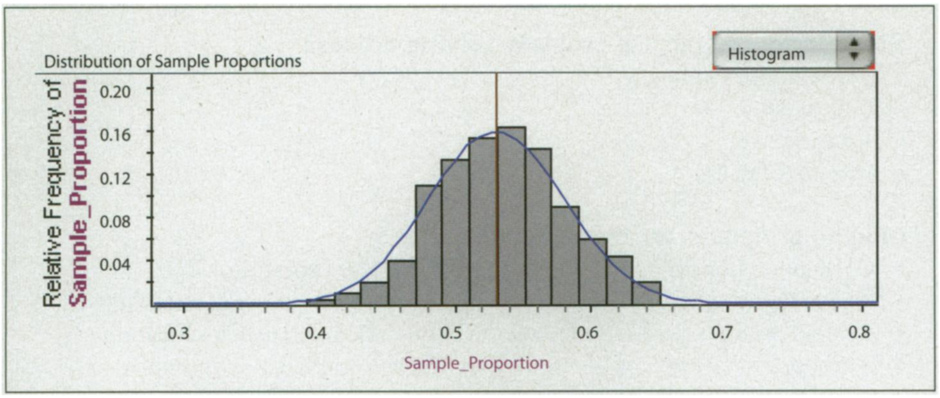

Fig. 2 Simulated sampling distribution for a sample proportion

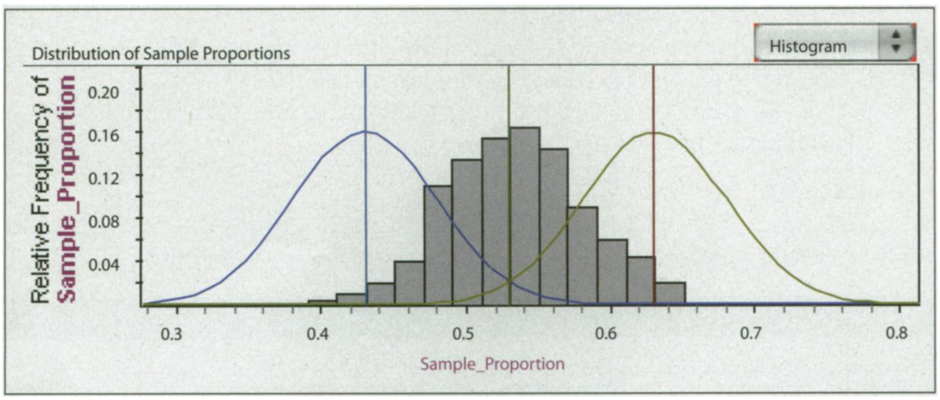

Fig. 3 Interval of plausible values for a population proportion

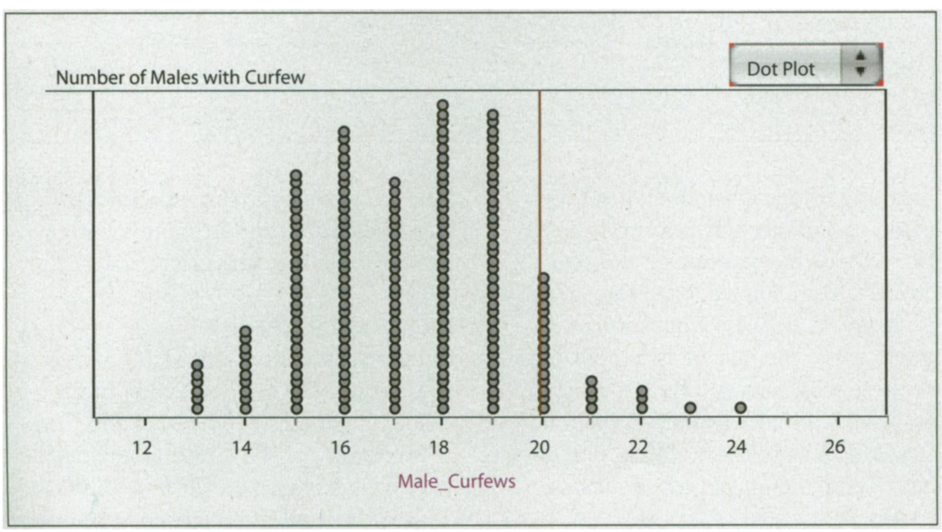

Fig. 4 Randomization distribution for the number of males with curfews

as 00 representing no. This sampling can be done by using a table of random digits, a computer (using software or Web sites such as www.random.org), or a graphing calculator (on a TI-83/84, use Math: Prb: 5RandInt(00,99)). **Figure 2** shows a simulated sampling distribution of sample proportions from 200 such samples. This sampling distribution has a mean of 0.53 and a standard deviation of 0.05 and is nicely represented by the normal distribution (overlaid smooth curve) with that same mean and standard deviation.

For a normal distribution, about 95 percent of measurements lie within two standard deviations of the mean. This measure of two standard deviations on either side of the mean, sometimes called the margin of error, forms an interval of reasonably likely outcomes. The margin of error is 2(0.05) = 0.10 for this example and would not change much even if the assumption of 0.53 for the true population proportion is off a bit. Finally, the plausible values of the population proportion that could produce such a sample outcome as reasonably likely must lie between 0.43 and 0.63 (0.53 ± 0.10), as displayed in **figure 3**. The two normal distributions sketched here are centered at 0.43 and 0.63, respectively, and have the same standard deviation (0.05). Each has the observed proportion of 0.53 at an edge of its reasonably likely outcomes. Thus, any normal distribution with nearly the same standard deviation that is located between these two distributions would also have the observed proportion of 0.53 within its reasonably likely outcomes. The interval of plausible values, 0.43 to 0.63, is called a 95 percent confidence interval.

One statistical analysis almost always leads to another. Because the observed proportion of the surveyed students having curfews was higher for males than for females, many students jumped to the conclusion that this proportion must be true for the school population as a whole. **Table 1** shows that the sample proportions of those having curfews were 20/32 = 0.625 for males and 33/68 = 0.485 for females. Remembering that sample proportions have inherent variation and that the sample size here is only 100, some students began to question whether this difference would hold up for the population of the whole school.

How can a statistical question that gets at this issue be formulated? The key lies in the randomness built into the sample selection process. Given that there are 68 females and 32 males in the sample and if we assume that a randomly selected male and a randomly selected female are equally likely to have a curfew, can the division seen in the observed sample (in which 33 females and 20 males have a curfew) be attributed simply to chance? This is a probability question that can be answered through simulation.

The simulation can be done manually or by using technology, but the principle is as follows. Gather 100 chips and mark 53 of them "curfew." Randomly split the 100 chips into piles of 68 "females" and 32 "males"; the randomization ensures that each "curfew" chip has an equal chance of ending up in either pile. Record the number of "curfews" among the "males." Repeat the process many times and plot the resulting distribution, now called a randomization distribution because it depends only on re-randomizing the observed data. **Figure 4** shows the result for one such simulation of 200 trials. Here, 20 or more males with curfews out of a sample of 32 males occurs 25 times out of 200 trials, or 12.5 percent of the time. In most statistical investigations, this would be considered fairly often, so there is little evidence to support a contention that the males in the school have a higher proportion of curfews. The split seen in the sample could have been due simply to the chance mechanism of random sampling. (*Teaching note*: Instead of chips, you can also use regular playing cards, colored beads, or any other objects of the same size that can be identified as curfew or noncurfew and randomly divided into two groups [male and female].)

Another question: Is the 68-to-32 split between females and males a bit unusual in a random sample of 100? This, too, can be verified by randomization.

A COMPARATIVE EXPERIMENT

The GAISE report and other curriculum guidelines state that high school students should understand the basic ideas behind both sample surveys and experiments. The following investigation illustrates some of the similarities and differences between surveys and experiments and incorporates an element of geometry as well.

The task is to approximate the area of a triangle. Each student in a class is asked to approximate the area through one of two methods: by using the area formula (area = [1/2] • base • height [method 1]) or by estimating through the use of graph paper (method 2).

One foundation of a good experiment is the random assignment of the methods (treatments) to students (experimental units). Assigning all the careful students to one method and all the careless ones to the other would *not* be the basis for a good experiment. Random assignment helps balance these characteristics (and any others that might affect the results) in the two treatment groups.

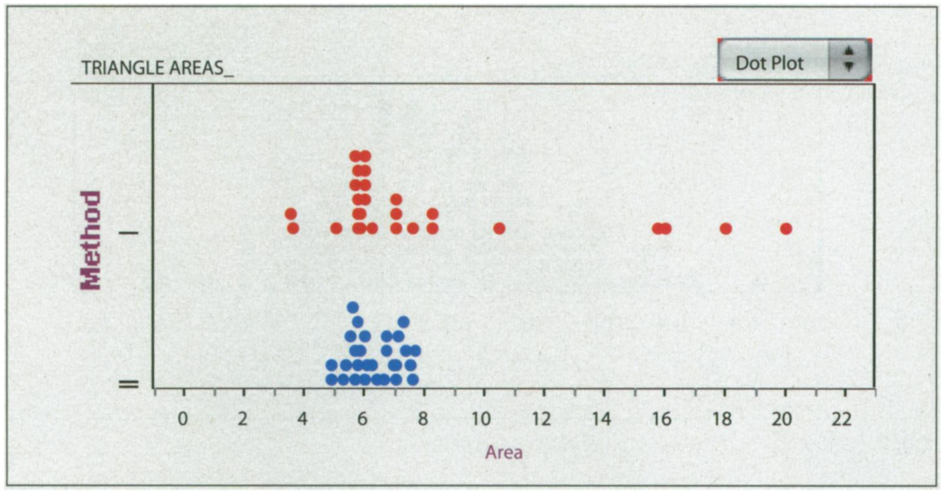

Fig. 5 Original area data (method 1 shown at top, method 2 shown at bottom)

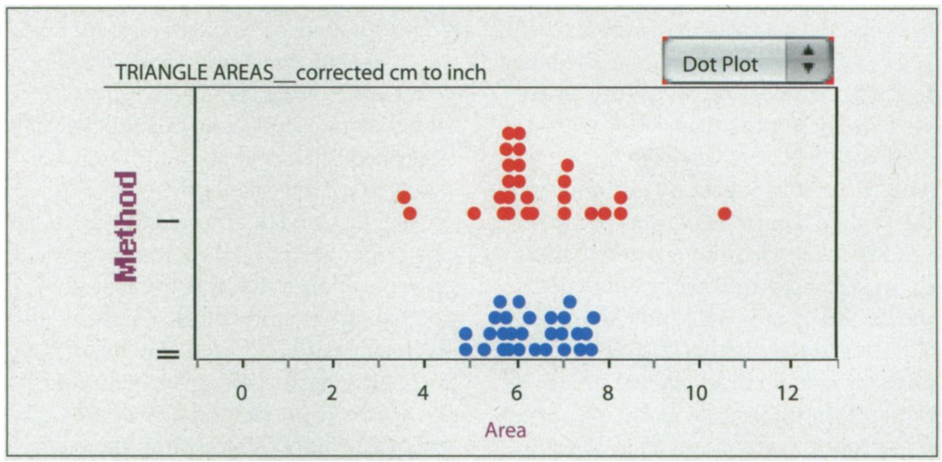

Fig. 6 Corrected area data (method 1 shown at top, method 2 shown at bottom)

Table 2

Summary Statistics for the Corrected Area Data

	Method 1	Method 2
Number	27.00	28.00
Mean	6.35	6.30
Median	6.00	6.16
Standard deviation	1.40	0.83

After assigning the treatments and collecting the data, have the students plot the data to see what they reveal. The original data are plotted in **figure 5**, from which it is quite obvious that four areas calculated by means of the formula are far from all the other calculations. Further checking revealed that these four students calculated in centimeters rather than inches; the corrected data are shown in **figure 6**. Being alert to outliers in a data set is strongly recommended, as these are often errors of measurement but on occasion may represent unique discoveries.

Class discussion focusing on shape, center, and spread (see **table 2** for summary statistics) should conclude that the two methods for computing the area of a triangle are about the same with respect to center but that the formula method may reflect greater variability. Even though one distribution is slightly skewed, using the sample mean as the measure of center is not unreasonable. The question on centers can then

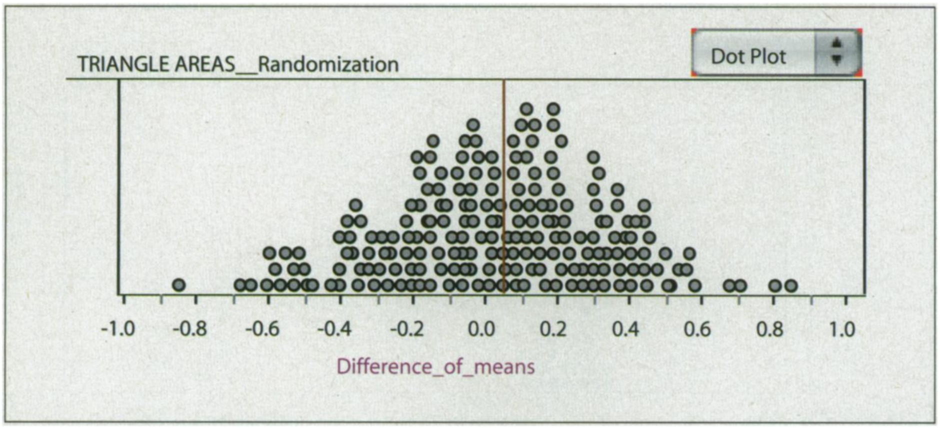

Fig. 7 Randomization distribution of difference in means

be phrased in terms of the difference between means: Can the difference between the sample means (observed to be about 0.05 square inch) be attributed to the randomization alone, or should we look for another cause?

If both methods worked equally well, we would expect no difference in the sample means. However, because randomization cannot guarantee that the two groups are exactly equivalent, the sample means will differ somewhat. To investigate whether the observed difference could be explained by the randomization process alone, we repeat that process many times so as to generate a simulated randomization distribution of the difference in sample means. If the observed difference is too big to be accounted for by chance alone, then a real difference in the methods is a more plausible explanation than is attributing the observed difference to the randomization.

Figure 7 provides a simulated randomization distribution of 200 trials each for the differences in means for the triangle area data. This distribution was produced by repeatedly splitting the 55 areas at random into groups of 27 for method 1 and 28 for method 2 and calculating the statistic of interest each time. The observed difference in means of 0.05 is almost exactly at the center of the rather symmetric distribution of the difference in means, as might have been expected. So this value could well be due to the students' randomization pattern rather than to any real difference in methods. (*Teaching note:* To perform this simulation by hand, students can write each of the 55 areas onto equal-sized slips of paper, mix them up, and divide them into two piles: 27 for method 1 and 28 for method 2. Alternatively, students could use the Randomization Distribution feature of CPMP-Tools software (www.wmich.edu/cpmp/CPMP-Tools/) described in Hart et al. (2007) or a TI-83/84 graphing calculator.

The fact that data from both methods center around 6.3 square inches is strong evidence that this number is a good approximation of the true area of the triangle. If a difference in means were evident, further investigation of the accuracy of the methods would be in order. The only difference in the methods, then, seems to be in the variation of the data, with method 1 being the more variable. With an appropriate measure of variation, such as the standard deviation or the mean absolute deviation as developed in Kader and Mamer (2008), comparisons of spreads can be made by randomization as well.

SUMMARY

Many interesting questions that can be posed in the classroom can be addressed through the methods presented in this article. For example, in a sample survey, students could investigate the question "Do boys or girls have more pairs of shoes?" In a comparative experiment, students could investigate the question "Does the amount of homework assigned have a significant effect on test results?" Even teachers can do their own research: "Do students who sit in the front of the classroom perform better than students who sit in the back?" In this last example (as in all experiments), random assignment of students is essential, because more motivated students may choose to sit in front.

Most often, statistics in high school mathematics is either embedded as part of algebra or set up as separate units in an integrated mathematics program. In either setting, statistics can vitalize the mathematics being taught because students tend to be inherently interested in data on practical problems relating to their lives and the world around them. Data distributions are built on real number lines and make use of proportional reasoning (e.g., relative frequency). Measures of center and spread can be formulated algebraically, and algebra can be used to show the effect of linear transformations on these measures (in changing from centimeters to inches, for example). The patterns that emerge in sampling and randomization distributions are the result of the laws of probability. However, the essentials of statistical reasoning must be the main focus in bringing statistics to life. And mathematics is a necessary underpinning to that reasoning.

ACKNOWLEDGMENTS

The authors thank Henry Kranendonk of the Milwaukee Public Schools, Milwaukee, Wisconsin, for the survey data on curfews and Tamara Lin of Wilson High School in Hacienda Heights, California, for helping collect data for the triangle experiment.

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